

PATIENT INFORMATION	SAMPLE DETAILS
Name: Mrs. Preethi Balasubramaniam	Sample ID: SMP-2024-990012
Age / Gender: 42 Years Female	Sample Type: Venous Blood (EDTA + Serum)
DOB: 22 August 1983	Collection: 04 Jun 2026 07:00 AM (Fasting 10 hr)
UHID: AIIMS-2024-056721	Received: 04 Jun 2026 08:15 AM
Ref. Doctor: Prof. Nikhil Tandon, MD DM – Endocrinology	Reported: 04 Jun 2026 01:30 PM

◆ GLYCEMIC PROFILE & HbA1c				Panel: GLY-001 Priority: ROUTINE
TEST NAME	VALUE	UNIT	REFERENCE RANGE	FLAG
Fasting Plasma Glucose (FPG)	182	mg/dL	70 – 99	HIGH ▲
Post-Prandial Glucose (2-hr PPG)	298	mg/dL	< 140	CRITICAL
HbA1c (Glycated Haemoglobin)	9.4	%	4.0 – 5.6 (Normal)	CRITICAL
Estimated Avg. Glucose (eAG)	221	mg/dL	70 – 126	HIGH ▲
Fructosamine	385	umol/L	205 – 285	HIGH ▲
Insulin (Fasting)	28.5	uIU/mL	2.0 – 25.0	HIGH ▲
C-Peptide	4.2	ng/mL	0.8 – 3.1	HIGH ▲
HOMA-IR	12.9		< 2.5	CRITICAL
HOMA-B (Beta Cell Function)	48	%	100 (Normal)	LOW ▼

◆ THYROID FUNCTION TEST (TFT)				Panel: THY-002
TEST NAME	VALUE	UNIT	REFERENCE RANGE	FLAG
TSH (Thyroid Stimulating Hormone)	7.82	uIU/mL	0.27 – 4.20	HIGH ▲
Free T4 (fT4)	0.72	ng/dL	0.93 – 1.70	LOW ▼
Free T3 (fT3)	2.8	pg/mL	2.0 – 4.4	Normal
Anti-TPO Antibodies	185	IU/mL	< 35	HIGH ▲
Anti-Thyroglobulin Ab	92	IU/mL	< 40	HIGH ▲

◆ METABOLIC & RENAL SCREEN				Panel: MET-003
TEST NAME	VALUE	UNIT	REFERENCE RANGE	FLAG
Serum Creatinine	1.05	mg/dL	0.5 – 1.1 (F)	Normal
eGFR (CKD-EPI formula)	68	mL/min/1.73m²	≥ 60	Normal
Urine Microalbumin / Creatinine	142	mg/g	< 30	HIGH ▲

Urine Microalbumin (24hr)	310	mg/day	< 30	HIGH ▲
Serum Sodium	136	mEq/L	136–145	Normal
Serum Potassium	4.1	mEq/L	3.5–5.1	Normal
Magnesium	1.6	mg/dL	1.7–2.2	LOW ▼
Vitamin D (25-OH)	14	ng/mL	30–100	LOW ▼
Vitamin B12	178	pg/mL	200–900	LOW ▼

◆ CLINICAL SUMMARY

Diagnosis: Type 2 Diabetes Mellitus – Poorly Controlled (HbA1c 9.4%)

1. **Severe Hyperglycaemia** — FPG 182, PPG 298, HbA1c 9.4%. Average blood glucose of 221 mg/dL over 3 months indicates inadequate glycaemic control. Intensification of OHA / insulin regimen strongly recommended.
2. **Insulin Resistance** — Fasting insulin 28.5 uIU/mL with HOMA-IR 12.9 (severe). Beta-cell exhaustion evident (HOMA-B 48%). GLP-1 agonist or SGLT-2 inhibitor addition advised.
3. **Early Diabetic Nephropathy** — Microalbuminuria 310 mg/day (Stage 2). Initiate ACE inhibitor / ARB. Renal function monitoring every 3 months.
4. **Hashimoto's Thyroiditis** — TSH 7.82 with elevated Anti-TPO (185 IU/mL) and low ft4 confirms hypothyroidism. Initiate levothyroxine 50 mcg OD, titrate in 6 weeks.
5. **Nutritional Deficiencies** — Vitamin D (14 ng/mL) and B12 (178 pg/mL) deficient. Supplement: Cholecalciferol 60,000 IU weekly x 8 weeks; Methylcobalamin 1500 mcg/day.

Prof. Nikhil Tandon, MD DM Head –
Endocrinology Dept. AIIMS New Delhi | MCI:
38901

Dr. Kavitha Shetty, MD Consultant – Clinical
Biochemistry AIIMS New Delhi | MCI: 52110

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